Project Title: Credit Risk Prediction & Default Analysis

Role: Machine Learning Analyst

1. Introduction & Business Context

In the banking sector, the primary risk is **Credit Default**—when a borrower fails to make required payments. This project aims to develop a predictive model using the credit.csv dataset to categorize loan applicants into "Good" (non-default) or "Bad" (default) risks.

Objective: To reduce the bank's "Bad Debt" ratio by identifying high-risk individuals before a loan is approved, while ensuring that "Good" customers are not unfairly rejected.

2. Dataset Overview

The dataset contains historical data on loan applicants with the following key features:

- **Target Variable:** Default (0 = Good loan, 1 = Defaulting loan).
- **Financial Indicators:** Checking_balance, Savings_balance, and Existing_credits.
- **Commitment Metrics:** Installment_rate (as a % of income) and Installment_plan.
- **Demographics:** Age, Personal_status (gender/marital status), and Dependents.

3. Methodology: Machine Learning Workflow

Step 1: Exploratory Data Analysis (EDA)

Analyze the dataset through statistics and visualizations to uncover patterns, detect outliers.

Step 2: Data Preprocessing

Clean the data by handling missing values, encode categorical text into numbers, and scale features to ensure the machine learning model can process the information accurately.

Step 3: Model Training

Evaluate and compare multiple classification algorithms (such as Logistic Regression, Decision Tree, Random Forest, and SVM) to identify the most accurate mathematical approach for predicting loan defaults.

4. Evaluation Metrics & Justification

For this business case, **Accuracy is not the primary metric.** Focus on:

- **Recall (Sensitivity):** High recall is vital because missing a "Default" (False Negative) costs the bank the entire principal amount of the loan.
- **F1-Score:** To balance the cost of losing a customer (False Positive) against the cost of a bad loan.
- **AUC-ROC:** To measure how well the model distinguishes between the two classes.

5. Business Impact & Recommendations

- **Model Selection:** Chose the best model based on evaluation metrics.
- **Risk Mitigation:** Reduce financial loses from unpaid loans.
- **Operational Efficiency:** How model can speed up the approval process.

6. Final Deliverable

1. Python Code File (.ipynb): Your code should be clean, commented, and follow the logical flow of a data science project.

2. PowerPoint Deck (The "Business" Presentation): This is where you "sell" your findings to the bank’s management.